Lesson 8.1 Guided Notes

Session 1 · Equations and Solutions

I can check whether a value is a solution of an equation by substituting it. · 6.EE.B.5

Word / Spanish	Meaning — complete the blank	Example
equation / ecuación	A math sentence that says two amounts are _____.	x + 3 = 10
variable / variable	A letter that stands for an _____ number.	the x in x + 3 = 10
solution / solución	A value that makes the equation _____.	x = 7
substitute / sustituir	_____ the variable with a number.	put 7 in place of x

Key idea

To test a value, _____ it for the variable and do the math.

If both sides are _____, the value is a solution. If not, it is not a solution.

Worked example

Is k = 7 a solution of k − 4 = 2? Is k = 6?

k = 7: 7 − 4 = _____. 3 ≠ 2, so the equation is _____.

k = 6: 6 − 4 = _____. 2 = 2, so the equation is true.

Answer: k = _____ is the solution.

You try

1. Substitute the value. Write T for true or F for false.
a. 4k = 20 (k = 5) _____
b. n − 2 = 6 (n = 4) _____

_____

2. Which of 6, 8 or 9 is the solution of 7m = 56?

_____

Sum it up: A number is a solution of an equation when ___ because ___.

_____

Session 2 · Writing Equations

I can write an equation for a situation and test a proposed answer. · 6.EE.B.6

Word / Spanish	Meaning — complete the blank	Example
variable / variable	A letter that stands for the _____ number.	let m = marbles lost
equation / ecuación	A math sentence with an _____ sign.	30 − m = 11
increased by / aumentado en	Means _____.	n + 9
decreased by / disminuido en	Means _____.	n − 9

Key idea

Name the unknown with a _____ and say what it means.

Write the story as a math sentence, like start _____ lost = left, or groups × size = _____.

Worked example

A jar had 30 marbles. Some rolled away and 11 are left. How many rolled away?

Let m = the marbles that _____.

Start − lost = left: 30 − m = _____.

Test m = 19: 30 − 19 = _____. True.

Answer: _____ marbles rolled away.

You try

1. Write an equation for each sentence. Use n.
a. A number plus 8 is 15. _____
b. 3 times a number is 21. _____

_____

2. A crate had 46 milk cartons. Some were taken, and 13 are left. Write an equation. Find the number taken.

_____

Sum it up: To write an equation for a story, I ___ because ___.

_____

Session 1 · Model, reason, and explain

Equations and Solutions · 6.EE.B.5

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Test x=6 and x=7 in x−3=4. Write both substituted statements and explain why only one is a solution.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: accepts a proposed value without testing the original equality.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Writing Equations · 6.EE.B.6

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A jar begins with 28 counters. Some are removed and 9 remain. Define r, write an equation, solve, and check the story.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: models the unknown as the remaining amount instead of the removed amount.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?